LEG-SEIR: Lagged Exposure for Graph-Based Dengue Forecasting

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***Abstract*—Recent local case counts provide a strong baseline for dengue forecasting, whereas useful signals from other districts may arrive with a delay. This paper proposes LEG-SEIR, a compact forecaster that combines a lagged-exposure graph with a Susceptible, Exposed, Infectious, Recovered (SEIR) model. A learned sparse graph and receiver-specific lag weights inform both a neural branch that preserves local data and a conservative SEIR decoder that handles imported pressure. A transition from historical to simulated pressure connects observed reports to the compartment model. A horizon-specific mixing method then combines persistence, local, graph, and mechanistic forecasts. The model is evaluated using source-audited Sri Lankan district reports, learning-rate selection on development data only, and a five-seed retrospective test on 2026 observations. LEG-SEIR achieves a mean count RMSE of 79.63, compared with 87.38 for persistence, 81.39 for an identity MLP, and 80.76 for a competitive shared-adapter control. Removing the graph or replacing the mechanistic branch increases the mean RMSE to 84.16 and 81.26, respectively. The improvement is strongest beyond the first forecast horizon. These findings support the complete architecture on this sample; however, they neither isolate each coupling mechanism nor establish prospective generalization. Frozen notebooks, source ledgers, predictions, and checkpoints are provided to support independent verification.**

***Index Terms*—dengue forecasting, graph neural networks, lagged exposure, SEIR, reproducibility**

I. Introduction

Dengue forecasts should be evaluated against simple surveillance-based baselines [1], [2]. Sri Lanka has experienced severe dengue outbreaks [3], and forecasting challenges highlight the need for comparable evaluation of epidemic predictions [2]. At the district level, a simple forecast repeats the latest reported count. This persistence baseline requires no training and, in the present evaluation, achieves an RMSE of 87.38. Several simple control models improve on this score (Table I). The central research question is therefore whether a coupled spatial and mechanistic model can achieve further improvement without sacrificing these competitive local predictors.

Lagged neighborhood signals have motivated spatial epidemic forecasting [4], [5]. The working hypothesis of this study is that the task can be improved by preserving a local predictor while selectively adding historical spatial information. A compartment decoder additionally requires an explicit rule for shifting from observed history to simulated states. These are design ideas tested through forecasting comparisons, rather than established biological transmission mechanisms.

This design question is investigated through the proposed Lagged-Exposure Graph-SEIR (LEG-SEIR) model. A compact shared multilayer perceptron (MLP) models local histories. A learned directed graph and specific lag weights reuse historical information in two ways: they feed neural graph messages and supply imported pressure within an SEIR rollout. A decaying historical anchor smoothly transitions to pressure derived from simulated neighbor states. A learned horizon-specific mixture retains persistence alongside the local, graph, and mechanistic outputs.

The complete architecture attains a lower retrospective 2026 count RMSE than persistence, an identity MLP, and a competitive shared-adapter baseline. The improvement over the shared adapter is modest, and the identity MLP remains competitive in development tests. The case study contains only 18 eligible origins and had previously informed the architectural design. Consequently, it constitutes a retrospective test rather than an untouched prospective test.

The contributions of this paper are as follows. First, a compact lagged-exposure coupling is introduced that connects a local-preserving graph encoder to an imported-pressure compartment rollout. Second, a multi-fold development and matched-control evaluation is presented that retains simple neural baselines and reports negative evidence. Third, a source-audited, independently verified experiment package is released. The claims are limited to a proposed design and an observed improvement; this work does not claim to be the first graph-compartment forecaster, nor to identify biological transmission routes.

II. Related Work

Spatial and local predictors. Graph WaveNet and MTGNN learn spatial dependencies [6], [7]. The static sparse receiver-sender graph adopted here follows this general adaptive-graph approach. STID demonstrates strong MLP forecasting with spatial and temporal identity [8], while simple linear models can rival complex time-series architectures [9]. These findings motivate the inclusion of local controls, even though their original tasks are not dengue comparisons.

Hybrid epidemic models. IeRNN combines epidemic recursion with neural neighboring effects [10]. Causal-

GNN couples graph learning with epidemic equations [11]. STAN and MPSTAN incorporate transmission-domain knowledge [12], [13]. MepoGNN shares a learned graph across neural and metapopulation modules while weighting mobility history [14]. HeatGNN models mechanistic heterogeneity [15], and M-Graphormer combines spatiotemporal learning with infectious disease dynamics [16]. Therefore, neither graph-compartment coupling nor historical weighting is entirely novel in isolation.

Dengue and lagged influence. Liu and Cao incorporate lagged neighborhood memory into multi-disease forecasting [4]. Weng et al. study Sri Lankan dengue graph forecasting with environmental inputs [17]; however, that work lacked an appropriate baseline and did not outperform the persistence baseline adopted here. It relied on a graph neural network without epidemiological knowledge such as SEIR dynamics. Liu et al. develop an explainable covariate compartmental model for Sri Lankan dengue [18]. Other studies employ the vector-host SEIR-SEI framework to explicitly represent mosquito compartments and epidemiological dynamics [19], [20]. Their tasks and inputs differ from the case-only comparison considered here. The proposed contribution is the specific reuse of case-lag weights in both neural messages and imported pressure, combined with a historical-to-simulated pressure transition and horizon mixing. The published mechanisms discussed in this section provide context but are not the reproduced competitors reported in the final results table.

Anchoring and evaluation. Persistence initialization in neural forecasting has been studied previously [21]. In LEG-SEIR, persistence is retained as a jointly trained mixture component. Forecasting evaluations require comparable targets [1], [2], and repeated selection can overfit finite validation data [22]. To mitigate this risk, learning rates are selected on pre-2026 folds, and the previously inspected case study is explicitly described as retrospective.

III. The LEG-SEIR Forecaster

A. Task and overall computation

At origin t, input $C_t \in \mathbb{R}^{N\times L}$ contains L observed reports for N districts, ending at $t-1$. We predict the next H reports, $Y_t \in \mathbb{R}^{N\times H}$. Columns $\ell = 1,\dots,L$ run from oldest to latest. Figure 1 shows the main dependencies. Raw counts initialize the decoder, while standardized logs supply the neural branches. A local representation is retained through a gated residual graph message. All four outputs are mixed before transformation back to counts.

Let $X = (\log(1+C)-\mu)/\sigma$, using training-only scalar statistics, and $p_{i,h} = X_{i,L}$ repeat the latest observation. The shared encoder ϕ reads levels and their consecutive differences:

$$h_i = \phi(X_i, \Delta X_i), \qquad f_i^L = W_o h_i + b_o. \tag{1}$$

The evaluated encoder has no district-ID input embedding. District-specific graph factors and lag parameters still condition the model; it is not an unseen-district inductive architecture.

B. Shared graph and historical lag selection

Learned receiver/sender factors $U, V \in \mathbb{R}^{N\times d_a}$ give scores $UV^\top$. Mask the diagonal, retain the largest K_g off-diagonal entries per receiver, and apply row-softmax to obtain A. Thus A_{ij} weights sender j for receiver i. The graph is learned during training and fixed during inference, rather than an input-dependent attention matrix.

Each receiving district has lag logits $\ell_i \in \mathbb{R}^L$, with $w_i = \mathrm{softmax}(\ell_i)$. The factorized operator $A_{ij}w_{i,l}$ separates spatial weighting from historical selection. The lag-weighted neural input and representation are:

$$q_{i,l} = L w_{i,l} \sum_j A_{ij} X_{j,l}. \tag{2}$$

$$\begin{aligned} h_i^G &= h_i + g_G W_m\big[\phi(q_i, \Delta q_i) - h_i\big], \\ g_G &= \mathrm{sigmoid}(a_G). \end{aligned} \tag{3}$$

The factor L makes uniform lag weights recover the ordinary graph-averaged history. The same encoder processes q, including its differences. Subtracting h_i forms a neighbour departure; gated addition retains the local path. The graph forecast reuses the local head: $f_i^G = W_o h_i^G + b_o$. Lag weights are recipient-specific, shared across senders and origins; they are not edge-specific incubation or travel delays.

C. Imported pressure and observation-to-simulation bridge

For fixed population P_i, let $K_i = P_i/11$ and $c^*_{i,l} = \mathrm{clip}(C_{i,l}/K_i, 10^{-6}, 0.5)$. The same graph and lag weights construct a historical imported-pressure anchor:

$$a_i = \sum_l w_{i,l} \sum_j A_{ij} c^*_{j,l}. \tag{4}$$

A rate head maps h_i^G to H logits $r_{i,h}$, converted to positive transmission proxies $\beta_{i,h}$. During solver day k, all districts have simulated infectious fractions $i_j(k)$. Historical pressure gradually gives way to these states:

$$\begin{aligned} n_i(k) &= \sum_j A_{ij} i_j(k), \\ d_k &= \exp(-k/\tau). \end{aligned} \tag{5}$$

$$P_i(k) = d_k a_i + (1-d_k) n_i(k). \tag{6}$$

$$\begin{aligned} \lambda_i(k) &= \beta_{i,h}\big[(1-\eta) i_i(k) + \eta P_i(k)\big], \\ \eta &= \mathrm{sigmoid}(a_\eta). \end{aligned} \tag{7}$$

Figure 2 separates the historical-to-simulated transition from local/import gating. At $k=0$, imported pressure uses the observed anchor; later steps increasingly use simulated neighbours. The decay constant is fixed. There is no assimilation of future reports during rollout.

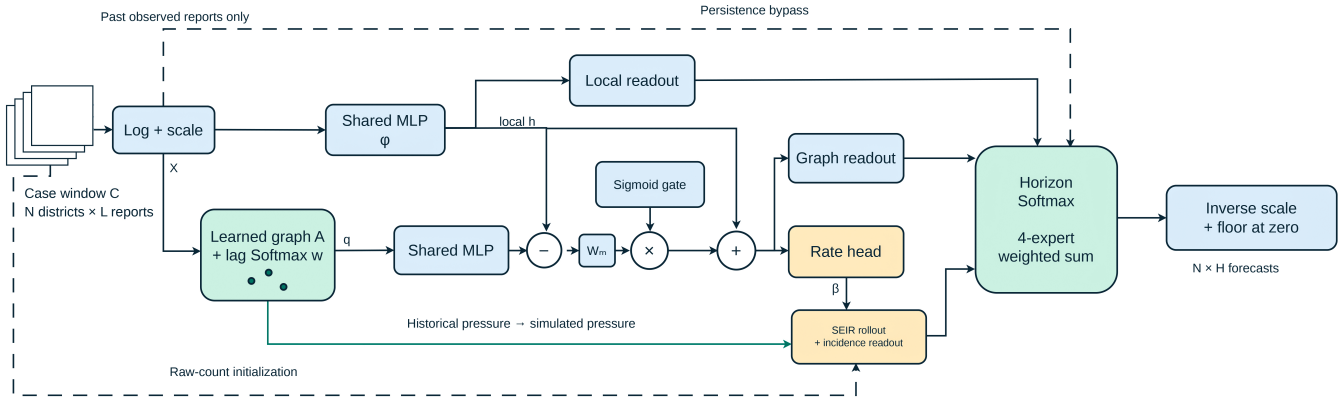


Fig. 1. LEG-SEIR architecture. Standard MLP/readout blocks are kept compact; green components highlight the proposed lag coupling and prediction fusion. The same graph and lag weights feed neural history aggregation and mechanistic imported pressure. Subtraction, linear projection, gating and addition preserve a local representation. Dashed paths carry raw counts and the persistence bypass. The local and graph readouts share parameters. Learned connections are predictive dependencies, not measured transmission routes.

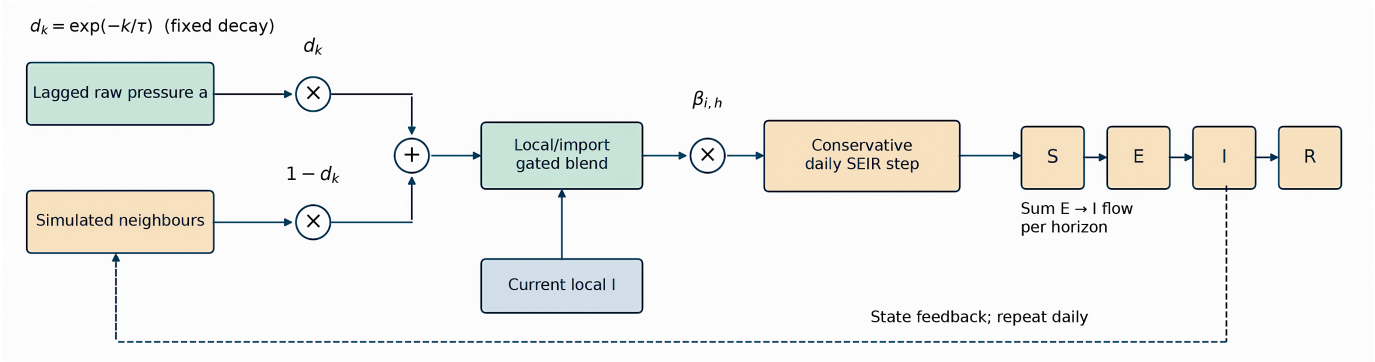


Fig. 2. Imported-pressure bridge. Historical and simulated neighbour pressure are weighted by d_k and $1 - d_k$, summed, then blended with local infectious pressure through a shared learned gate. The neural rate supplies infection force to a conservative SEIR-inspired step. Simulated infectious states feed the next daily step; exposed-to-infectious flows provide the incidence-style readout. This schematic describes computation, not validated disease mechanisms.

D. Conservative incidence decoder

SEIR partitions a population into susceptible, exposed, infectious and recovered states [23]. Here normalized fractions are *proxies*. At each origin, initialize $s = 0.318$, $e = c^*_{i,L-1}$, $i = c^*_{i,L}$, and $r = \max(0, 1-s-e-i)$, then normalize to unit mass. Each daily step computes bounded pre-update flows:

$$u = s(1 - e^{-\lambda}), \qquad v = e(1 - e^{-\omega}), \qquad z = i(1 - e^{-\gamma}). \tag{8}$$

$$s^+ = s - u, \qquad e^+ = e + u - v, \qquad i^+ = i + v - z, \quad r^+ = r + z. \tag{9}$$

Every outflow is bounded by its source compartment, preserving nonnegativity and total mass. Sum v over D daily steps per horizon and scale by K_i to produce $y^S_{i,h}$. Transform this output to standardized log space f^S. Numerical conservation does not validate biological states, and the decoder does not explicitly represent mosquitoes, serotypes or reinfection. Vector-host SEIR-SEI models explicitly represent mosquito compartments [19], [20].

E. Horizon mixing and learning objective

Learned horizon logits $\theta \in \mathbb{R}^{H\times 4}$ yield four softmax weights, shared across districts and origins:

$$\widehat{z}_{i,h} = \alpha_{h,p} p_i + \alpha_{h,L} f^L_{i,h} + \alpha_{h,G} f^G_{i,h} + \alpha_{h,S} f^S_{i,h}. \tag{10}$$

Inverse transformation gives $\widehat{y} = \max(0, \exp(\mu + \sigma\widehat{z}) - 1)$. Mixing occurs in log space, not count space. The head learns a separate four-expert mixture for each horizon. Experts learn jointly, so mixture weights do not identify causal contributions or the quality of separately trained predictors.

For training origins $\mathcal{T}$, the objective is

$$\mathcal{L} = \frac{1}{|\mathcal{T}|NH} \sum_{t\in\mathcal{T},i,h} (\widehat{z}_{t,i,h} - z_{t,i,h})^2. \tag{11}$$

Checkpoint selection instead minimizes validation count RMSE; the two criteria emphasize different errors.

IV. Data and Experimental Protocol

A. Source reconstruction and eligible windows

We reconstruct counts from Sri Lanka's Epidemiology Unit Weekly Epidemiological Reports[1] and use Census 2012 population tables[2]. The curated source snapshot is pinned by its manifest. The source manifest contains 714 entries, including 685 original report PDFs across 686 report slots through Vol. 53 No. 33. Aggregating Ampara and Kalmunai maps 26 reporting divisions to 25 districts. Only case histories and fixed population scaling enter the model; no mobility, boundaries or weather are inputs.

The frozen source policy accepts coherent printed counts and documented arithmetic repairs, retaining unresolved reports as missing. Identical coherent cumulative counts accompanying repeated positive weekly counts cannot certify zero incidence. No count is removed because of its forecast error. Incomplete input/target windows are excluded; the supplement records parsing, repairs and coverage. Code is maintained in the project repository and the curated dataset is intended for Hugging Face release.

Targets use report-period year, not recovered release vintages. Forecast horizons are subsequent report observations, whereas the decoder assumes seven daily steps per horizon. This mismatch and retrospective source reconstruction limit operational interpretation. Final splits contain 525 training origins through 2024, 40 validation origins in 2025 and 18 test origins in 2026. All compared methods use identical eligible target windows; earlier observed 2026 reports may enter later rolling test inputs, but test targets never update parameters.

B. Development selection and final fits

A pre-2026 validation screen shortlisted three families: horizon mixture, lagged exposure and compartment-only observer. Development A trains through 2022 and validates 2023; B trains through 2024 and validates 2025. Fifteen configurations, two learning rates and three seeds in both folds give 180 fits. Each configuration selects the LR maximizing mean fold skill $1 - \mathrm{RMSE}/\mathrm{RMSE}_{\mathrm{persistence}}$; ties favour the lower LR. Matched removals retain their parent's LR.

Final training through 2024 selects checkpoints on 2025 and scores 2026 using five seeds. Seventy-five primary fits and ten additional horizon-removal LR sensitivities bring the total to 265 neural fits, plus three fixed-alpha ridge fits. Selection seeds 0-2 overlap final seeds 0-4. The supplement provides the chronology and selection records. LEG-SEIR was shortlisted before final fitting, but 2026 had previously influenced research ideas; this comparison is retrospective.

C. Controls and implementation

Controls include persistence, shared ridge ($\alpha = 1$; five level/difference inputs), local MLP, identity MLP, plain graph, local SEIR, uniform mixture, shared adapter, horizon mixture and compartment-only observer. The identity MLP adds a learned four-dimensional district embedding. The shared adapter uses a rank-eight correction with its adapter parameters exempted from weight decay. Complete wiring and capacities are supplied; matched training recipes do not imply identical parameter counts.

LEG-SEIR graph removal disables both neural graph messages and imported pressure; its mechanistic replacement retains lagged neural messages but replaces the SEIR output with a direct neural branch. These controls do not isolate imported pressure or the decay transition individually. Local SEIR uses the inherited two-stage anchored blend, whereas graph-off uses a four-expert mixture; they are distinct controls.

In the evaluated implementation, $N = 25$, $L = H = 3$, width is 64, $d_a = 8$, $K_g = 5$, $D = 7$, $\tau = 7$, $\omega = 0.1$ and $\gamma = 1/7$. Transmission is $\beta = \exp[\mathrm{clip}(r - 1.1, -9, 1.1)]$. Graph/import gate logits initialize at -2; lag and mixture logits at zero. LEG-SEIR has 9,519 parameters versus identity MLP's 5,095. Adam [24] uses batch 128, weight decay 10^{-4}, gradient clipping at 5, at most 300 epochs and patience 40. LEG-SEIR's selected LR is 0.003; identity MLP and shared adapter use 0.001. Runtime versions and source hashes are in the supplement and execution archive.

D. Metrics and verification

Count RMSE pools origins, districts and horizons equally:

$$\mathrm{RMSE} = \sqrt{\frac{1}{ONH}\sum_{t,i,h}(\widehat{y}_{t,i,h} - y_{t,i,h})^2}. \tag{12}$$

We report the mean and sample SD of five seed-specific RMSEs, not ensemble RMSE or a confidence interval. MAE and sMAPE are secondary; percentage measures require care near zero [25]. With 18 overlapping origins, the 1,350 scored cells are not independent samples; dependence also matters for resampling uncertainty [26]. No significance claim is made. Verification checks 915 hashes, all 265 neural CPU reloads, reconstructed metrics, source truths, scaling, chronology and LR selection; it does not refit ridge.

V. Results

A. Observed improvement and competitive controls

Table I and Fig. 3 compare the primary sample. LEG-SEIR reaches 79.63 ± 0.64 RMSE, 8.87% below persistence and 2.16% below identity MLP. Shared adapter reaches 80.76 ± 0.38; LEG-SEIR's relative RMSE reduction against it is only 1.39%. This competitive result must remain visible. LEG-SEIR also has lower mean MAE than identity MLP, but not every secondary metric favours it over every removal.

B. Development evidence and horizon dependence

At the chosen rates, LEG-SEIR has mean development RMSE 50.86 in 2023 versus identity MLP's 51.17. In 2025, identity MLP slightly leads: 19.32 versus 19.34. These results constrain any general claim of graph superiority. Final horizon RMSEs for LEG-SEIR are 41.83, 80.52 and 103.87, compared

[1] https://www.epid.gov.lk/weekly-epidemiological-report/weekly-epidemiological-report

[2] https://www.statistics.gov.lk/

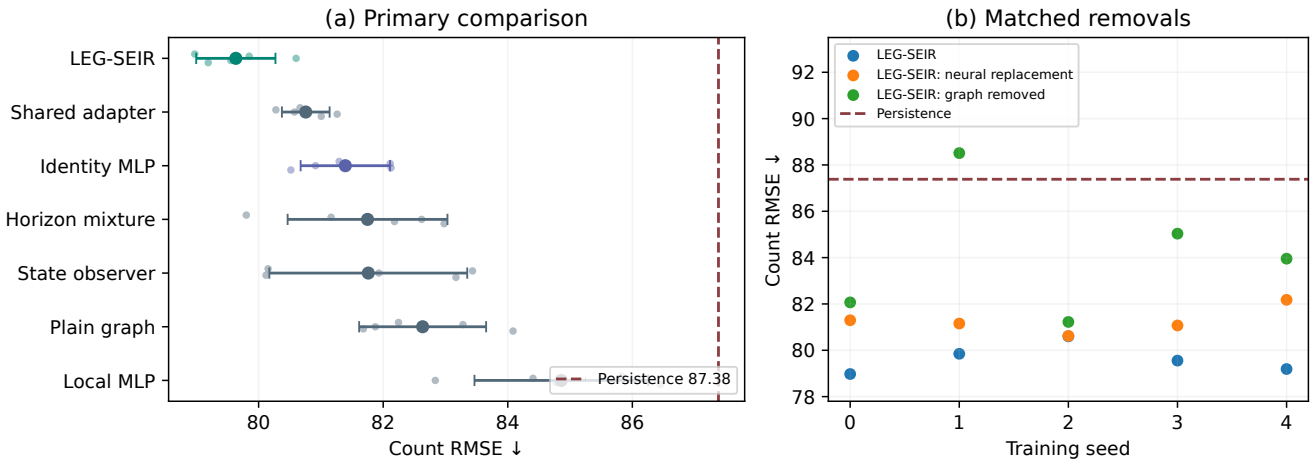


Fig. 3. Primary comparison and matched removals. (a) Large points show mean RMSE, bars show training-seed SD and small points show individual seeds; bars are not confidence intervals. (b) The complete model beats both removal controls in each seed, but the seed-two mechanistic difference is small. Graph removal jointly changes neural messages and imported pressure.

TABLE I
PRIMARY 2026 RESULTS. NEURAL RMSE: MEAN ± SAMPLE SD OVER FIVE SEEDS; MAE: SEED MEAN. SAME 18 ORIGINS FOR ALL ROWS.

Method	RMSE	MAE
Persistence	87.38	34.69
Shared ridge	85.54	34.93
Local MLP	84.86 ± 1.39	33.80
Identity MLP	81.39 ± 0.72	32.50
Plain graph	82.63 ± 1.02	33.23
Local SEIR	84.89 ± 3.17	33.72
Uniform mixture	83.15 ± 4.17	33.19
Shared adapter	80.76 ± 0.38	32.45
Horizon mixture	81.75 ± 1.28	32.68
Compartment observer	81.76 ± 1.59	32.88
LEG-SEIR (proposed)	$\mathbf{79.63 \pm 0.64}$	**32.14**
LEG-SEIR: neural replacement	81.26 ± 0.57	32.87
LEG-SEIR: graph removed	84.16 ± 2.86	33.52

with persistence at 42.63, 89.22 and 114.58 (Fig. 4). Most improvement occurs beyond the first horizon; not every seed improves first-horizon error.

C. What the component comparisons establish

Graph removal gives 84.16 RMSE; mechanistic replacement gives 81.26. The complete model wins both comparisons in all five seeds, but the seed-two difference against replacement is approximately 0.01848 RMSE. This supports the complete configuration on these observations, not a separately established advantage for each pressure mechanism. The original no-import screen had mixed seed effects and used a smaller budget.

SEIR is not universally helpful: the ordinary horizon mixture scores 81.75, whereas its direct replacement scores 81.36 at the parent's LR. The compartment-only observer reaches 81.76. Additional horizon-removal LR sensitivities and all negative results are retained in the supplement. The final comparison evaluates three shortlisted families and controls, not all nine screened families.

Supplementary national-total curves use five-seed ensembles rather than district-pooled seed scores. They still lag some rapid growth and overpredict later observations; lower pooled RMSE does not imply accurate prediction of every outbreak phase.

VI. DISCUSSION AND CONCLUSION

The findings support the use of this compact, coupled forecasting model on the retrospective dataset. The main scientific goal was to exploit historical spatial data without losing the strong local predictor, and then to shift this information into simulated infection pressure for longer-horizon forecasts. The model achieved lower errors than both the identity MLP and the removal controls. This result is consistent with the motivation, although it does not pinpoint exactly why the improvement occurred. Since shared graph modeling, lagged neighborhood effects, and mechanistic neural networks already exist, the primary contribution lies in the specific manner in which these components are combined (coupled), rather than in the invention of the individual parts.

Several limitations remain. First, the evaluation covers only one country with 18 overlapping origin points, and the 2026 data influenced earlier design choices. Second, because the exact historical report release dates could not be recovered, the data reconstruction is retrospective. Third, the alternative policy of simply repeating printed reports was not tested. Fourth, the SEIR constants and the seven-day solver mapping are assumptions that require further sensitivity analysis. In addition, the human-only compartment model does not explicitly account for dengue mosquito (vector) dynamics. Finally, removing the graph does not fully isolate the effect of imported pressure, and externally published hybrid models were not rerun as matched baselines.

Future research should compare uniform lag weights against learned lag weights. It should also isolate the import-gating mechanism while keeping the neural messages active, test the

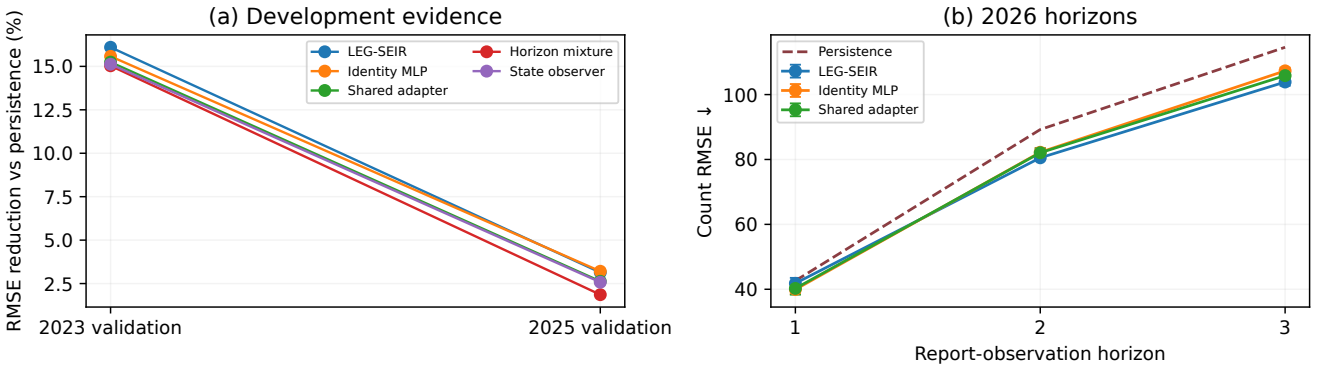


Fig. 4. Development and horizon evidence. (a) Fold-specific skill at development-selected learning rates; identity MLP remains competitive, especially in 2025. (b) Five-seed mean count RMSE by horizon, with seed SD. The proposed model's larger gains occur at horizons two and three.

removal of the history-to-simulation transition, and compare separate coupling operators against shared ones. Crucially, future experiments must make development decisions and evaluate the model on an independent, future dataset rather than repeatedly optimizing on the already-seen 2026 sample. For real-world operational use, the model will also need to incorporate true release-vintage data and provide calibrated uncertainty estimates.

In conclusion, LEG-SEIR reduces the mean retrospective RMSE relative to strong, simple baselines while maintaining a compact and explicitly defined computational structure. At this stage, the model serves as a reproducible architectural hypothesis supported by comparative evidence. However, independent confirmation and targeted ablation studies are necessary before stronger claims can be made about the underlying biological mechanisms or the model's ability to generalize to new regions.

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